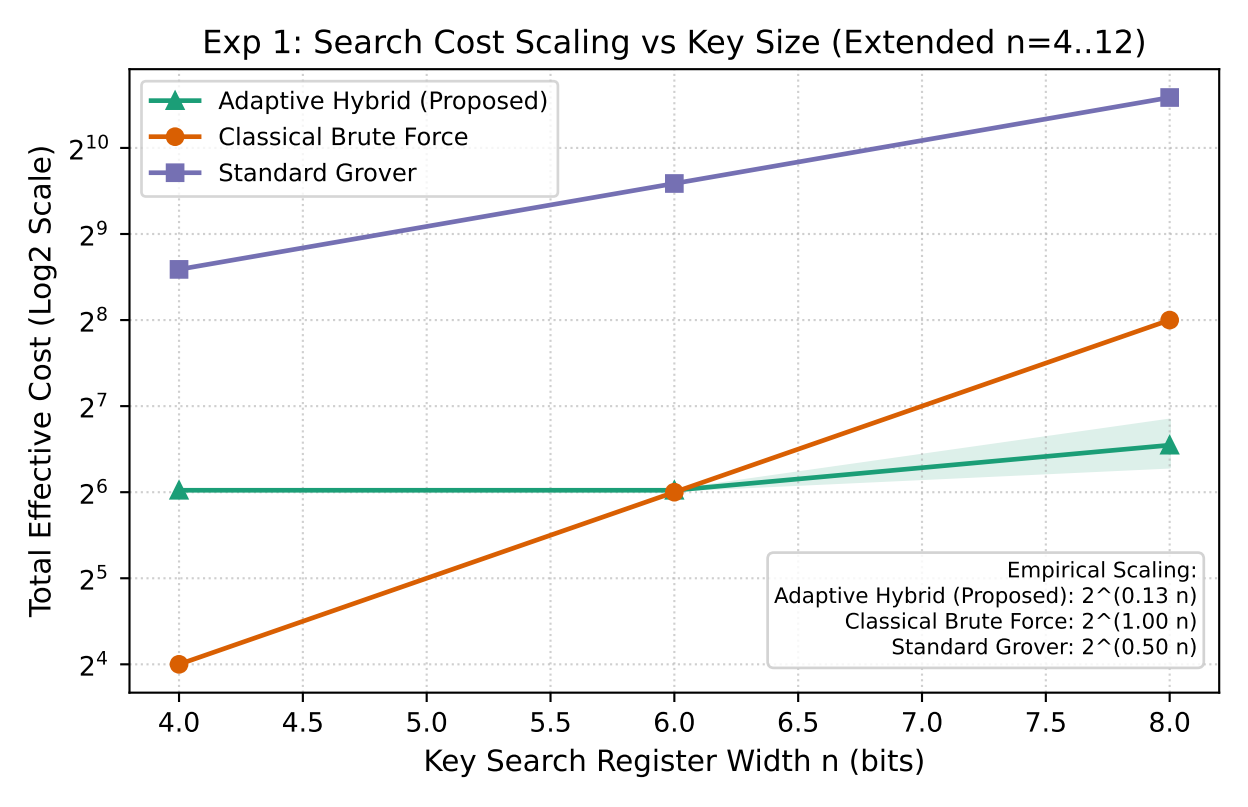
Exp 1: Search Cost Scaling vs Key Size (Extended n=4..12)
Adaptive Hybrid (Proposed)
Classical Brute Force
Standard Grover
Total Effective Cost (Log2 Scale)
2^{10}
2^{9}
2^{8}
2^{7}
2^{6}
2^{5}
2^{4}
Empirical Scaling:
Adaptive Hybrid (Proposed): 2^(0.13 n)
Classical Brute Force: 2^(1.00 n)
Standard Grover: 2^(0.50 n)
4.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
Key Search Register Width n (bits)